L46 Project

Exploring Quantisation In FPGA-Accelerated Binarized Neural Networks

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1 Introduction

Talk about quantisation as a method to speed up neural network accel, week 7 comp arch presentation was quite useful for this.

2 System Setup

To evaluate neural network acceleration on FPGA hardware, the FINN framework was deployed on a Diligent Arty-Z7-20 development board. FINN is an open-source framework developed by Xilinx/AMD that enables deployment of quantized neural networkks (QNNs) on FPGA platforms through automated hardware generation and compilation.

2.1 Hardware Platform

The target platform used in this project is the Arty-Z7-20 development board, this board includes an Artix-7 FPGA as the programmable logic (PL) and an ARM Cortex-A9 processor in the form of a Zynq-7000 SOC as the programmable system (PL).

2.2 Software Environment

- PYNQ - A pre-built PYNQ (Linux for Zynq SOCs which interfaces with PL using Python) image was flashed onto a microSD card and inserted onto the board.
  - There are a small number of pre-built images which AMD maintains, the Arty-Z7-20 isn't one of these boards. Luckily, one of the supported boards (Pynq-Z1), happens to have the exact same specs and (almost the same) pinouts as the Arty-Z7-20, we use the PYNQ v3.0.1 image in this work
- FINN - A compiler which maps Quantized Neural Networks (QNNs) trained in Brevitas to dataflow-style FPGA architectures [1].
- Xilinx Design Suite 2022.2

Figure 1: Image of the Arty-Z7-20

2.3

2.4

3 Experiment 1: Impact of quantisation on neural network FPGA performance

4 Experiment 2: Investigating parallelism in FPGA QNN performance

References

[1] Yaman Umuroglu, Nicholas J. Fraser, Giulio Gambardella, Michaela Blott, Philip Leong, Magnus Jahre, and Kees Vissers. FINN: A Framework for Fast, Scalable Binarized Neural Network Inference. In *Proceedings of the 2017 ACM/SIGDA International Symposium on Field-Programmable Gate Arrays*, pages 65–74, February 2017. `arXiv:1612.07119`, `doi:10.1145/3020078.3021744`.